

SALES GARUT INTELLIGENCE PLATFORM

Product Requirements Document - v0.2

Validated Build Specification

Initial Deployment	DSO Garut
Primary Users	DSM/Admin, SPV, Salesman
Primary Platform	Responsive Web Application
Data Entry	Excel Upload + Manual Input
Initial Data Period	2025-2026 running
Status	Development Ready; Production Data Gate remains

Purpose: transform Sales Garut operations from spreadsheet-dependent monitoring into one operational sales intelligence platform.

1. Document Purpose & Decision Status

This PRD is the primary product specification for the first build of Sales Garut Intelligence Platform. It consolidates the original product vision, the Antigravity discovery/audit findings, and the business decisions that have since been validated. It is intentionally detailed enough to build from, while leaving unverified business rules explicitly open rather than guessing them.

Build philosophy: preserve useful business logic from Excel, but do not clone the workbook. Build the best operational product experience for the underlying business decisions.

Area	Status
Product requirements	LOCKED unless explicitly marked TBD
Data architecture direction	LOCKED at conceptual level; implementation may refine names/indices
UI pixel design	NOT locked; must follow UX principles and module outcomes
AR incentive formula	TBD - do not infer
Full transaction history	Required before production UAT
GPS visit tracking	Out of V1; future architecture only

2. Product Vision

Sales Garut Intelligence Platform is an end-to-end operational sales intelligence platform that connects sales, target, products, outlets, territory, sales force, KPI, incentive, stock, and accounts receivable into one easy-to-use system.

The product must progress from monitoring to explanation to action:

MONITORING -> ANALYSIS -> INSIGHT -> ACTION

North Star: Make every important sales decision visible, measurable, explainable, and actionable.

3. Product Principles, Goals & Non-Goals

3.1 Product Principles

- Simple outside, intelligent inside - non-technical users should not need to understand databases, formulas, or BI tooling.
- Business platform, not dashboard only - data ingestion, master data, metrics, drilldown, alerts, and actions are part of the product.
- One source of truth - identical filter scope must produce identical metric values across modules.
- Explainable - every insight must be traceable to underlying data and filters.
- Configurable where business rules can change - calendars, thresholds, KPI rules, and program definitions must not be hard-coded.
- Expandable - architecture should not hard-code Garut so deeply that a future district requires a rebuild.

3.2 Primary Product Goals

- Know current sales performance and whether the business is on pace to target.
- Identify which salesman, product, outlet, rayon, or kecamatan is driving a result.
- Measure registered outlet vs active outlet coverage and find whitespace opportunities.

- Understand outlet contribution, recency, frequency, fluctuation, and product mix.
- Monitor Must Have and NPL penetration/repeat performance.
- Turn KPI and incentive calculations into a transparent, configurable system.
- Read current stock risk and accounts-receivable risk alongside commercial performance.
- Allow team members to update data safely through Excel upload and authorized manual input.

3.3 Non-Goals for V1

- Live GPS tracking
- GPS geofence check-in
- Automatic route optimization
- Machine-learning forecasts
- Automatic ERP sync
- Full warehouse management / inventory movement system
- Native iOS/Android app

4. Current Data Reality & Production Gate

Discovery confirmed useful operational logic and multiple structured data sources, but also found important quality constraints. The current transaction extract is a sampled development dataset, not a complete production history. Current customer data also contains multiple source-code formats and duplicate representations. Therefore, the architecture must treat raw source identifiers as aliases and use a canonical internal outlet identity.

Domain	Current Reality	Product Treatment
Transactions	Sampled historical extract; suitable for development, not production truth	Full unsampled extract required for final UAT
Customer / CL	Complete universe recoverable, but multiple customer-code formats and duplicate representations exist	Canonical outlet ID + alias mapping required
Target	Operational quantity target in cartons exists	Use as primary sales target
Stock	Point-in-time snapshot exists	V1 supports snapshots; no movement ledger yet
AR	Invoice/outstanding data available in discovery assets	Use official recurring AR export for production
GPS / Visits	No current visit/GPS fact data	Future only

Development may proceed now. Production sign-off requires full transaction data to pass reconciliation and UAT.

5. Users, Roles & Business Structure

5.1 Roles

Role	Default Scope
DSM / Admin	All Garut data; Data Center; masters; targets; KPI/incentive config; stock; AR; users; settings
SPV	Assigned team, salesmen, outlets, territory, KPI, sales performance; input/upload only if granted
Salesman	Own assigned business scope, target, outlets, KPI/incentive,

	performance, relevant product/stock info
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Permissions must be capability-based where practical (for example can_upload_sales, can_edit_customer, can_edit_target, can_manage_incentive) rather than assuming every user in the same role has identical edit rights.

5.2 Validated Organizational & Geographic Model

Organization: DSO -> Sub-DSO -> SPV -> Salesman -> Rayon -> Outlet | Geography: Kecamatan -> Outlet

- Sektor is retired / not used in V1.
- Kecamatan is an independent geographic dimension, not an exclusive salesman territory.
- One kecamatan may contain outlets owned by multiple salesmen.
- Rayon represents the operational beat dimension and is part of the current sales structure.

6. Core Business Rules & Metric Definitions

Metric	Definition	Notes
Primary Sales Target	Quantity / Cartons (KTN)	Official operational target framework.
Sales Value	Sales Netto / financial value	Analytical metric; not the primary sales quota.
Active Outlet (OC)	Distinct registered outlets with ≥ 1 qualifying purchase in selected calendar month	Monthly activity metric.
Registered Outlet (CL)	Registered outlets within the selected business/filter universe	Coverage denominator.
Coverage %	$\text{Active Outlet} / \text{Registered Outlet} \times 100\%$	Must work at DSO, SPV, salesman, rayon, kecamatan and product-program scopes where appropriate.
Dormant Outlet	No purchase for ≥ 60 consecutive days	Default threshold configurable.
Time Rate	HKE / HK	Working-day progress.
Achievement %	Actual / Target	Carton performance unless component defines another unit.
Remaining Target	$\max(\text{Target} - \text{Actual}, 0)$	Avoid confusing negative gap display in UI.
Required Daily Pace	Remaining Target / Remaining HK	Only when Remaining HK > 0 .
Latest Estimate (LE)	Actual / HKE $\times$ HK	Only when HKE > 0 .
Outlet Contribution	Outlet sales / total sales under current filters	Show quantity and value contribution.
Stock Cover - Target	Available Stock / (Monthly Target / HK)	Target-run-rate perspective.
Stock Cover - Actual	Available Stock / average actual daily sales over configured rolling period	Actual-velocity perspective.

6.1 Salesman Ownership Attribution

Management reporting follows the current owner. If an outlet is reassigned, historical outlet sales are attributed to the new/current owner for management views. However, transaction audit must preserve the original invoice salesman. Store both invoice_salesman_id and current_owner_salesman_id (derived from current assignment).

7. Product Programs, KPI & Incentive Rules

7.1 Product Hierarchy

Principal / Category / Brand / Group SKU / Sub-brand -> SKU

Hierarchy and mappings must be master-data driven so future SKU, brand, NPL, or principal additions do not require source-code changes.

7.2 Must Have Penetration

Must Have Line	Target Penetration
KTG	45%
DELI	45%
RTD	30%
FOX	35%
UHT	40%

For each line and scope: Target OC = Registered Outlet (CL) x line penetration target. Actual OC = distinct CL outlets purchasing the mapped Must Have line during the selected month. One outlet counts at most once within a line, but can count in multiple lines. Total Must Have Achievement = $\text{sum}(\text{Actual OC across lines}) / \text{sum}(\text{Target OC across lines})$. Thresholds must be configurable.

7.3 NPL Program

- NPL programs must be configurable with product/SKU mapping, launch date, campaign period, M1 target, M3 target and active status.
- Current known targets: M1 penetration 70%; M3 penetration 85%.
- RO-1: outlet with exactly 2 campaign orders.
- RO-2+: outlet with 3 or more campaign orders.
- Metrics: CL, NPL OC, penetration, first-order drop size, RO-1, RO-1%, RO-2+, RO-2+%, repeat drop size and drop-size movement.

7.4 Incentive Components

#	Component	Mangkok	Min	Max	Rule
1	Realisasi vs Target Qty - Kopi	Rp300,000	80%	150%	Proportional above minimum; capped
2	Realisasi vs Target Qty - Beverage	Rp200,000	40%	150%	Proportional above minimum; capped
3	Realisasi vs Target Qty - Non Kopi & Non Bvg	Rp200,000	85%	150%	Proportional above minimum; capped
4	Realisasi vs Target Value - All	Rp600,000	80%	150%	Dedicated incentive value target
5	OC Must Have SKU	Rp750,000	25%	25%	Binary full payout once threshold reached
6	AR Performance	Rp200,000	70%	100%	Thresholds locked; exact performance formula TBD

Base Mangkok Rp2.25M. Components 1-4: Rp0 below minimum; otherwise Mangkok x Achievement capped at Max. Component 5: full Rp750k at $\geq 25\%$. Component 6: AR payout formula TBD.

8. Conceptual ERD & Data Architecture

The ERD below is a build-level conceptual model. Table names may be refined during implementation, but the separation of raw identity, canonical dimensions, facts, business rules, and audit/governance must be preserved.

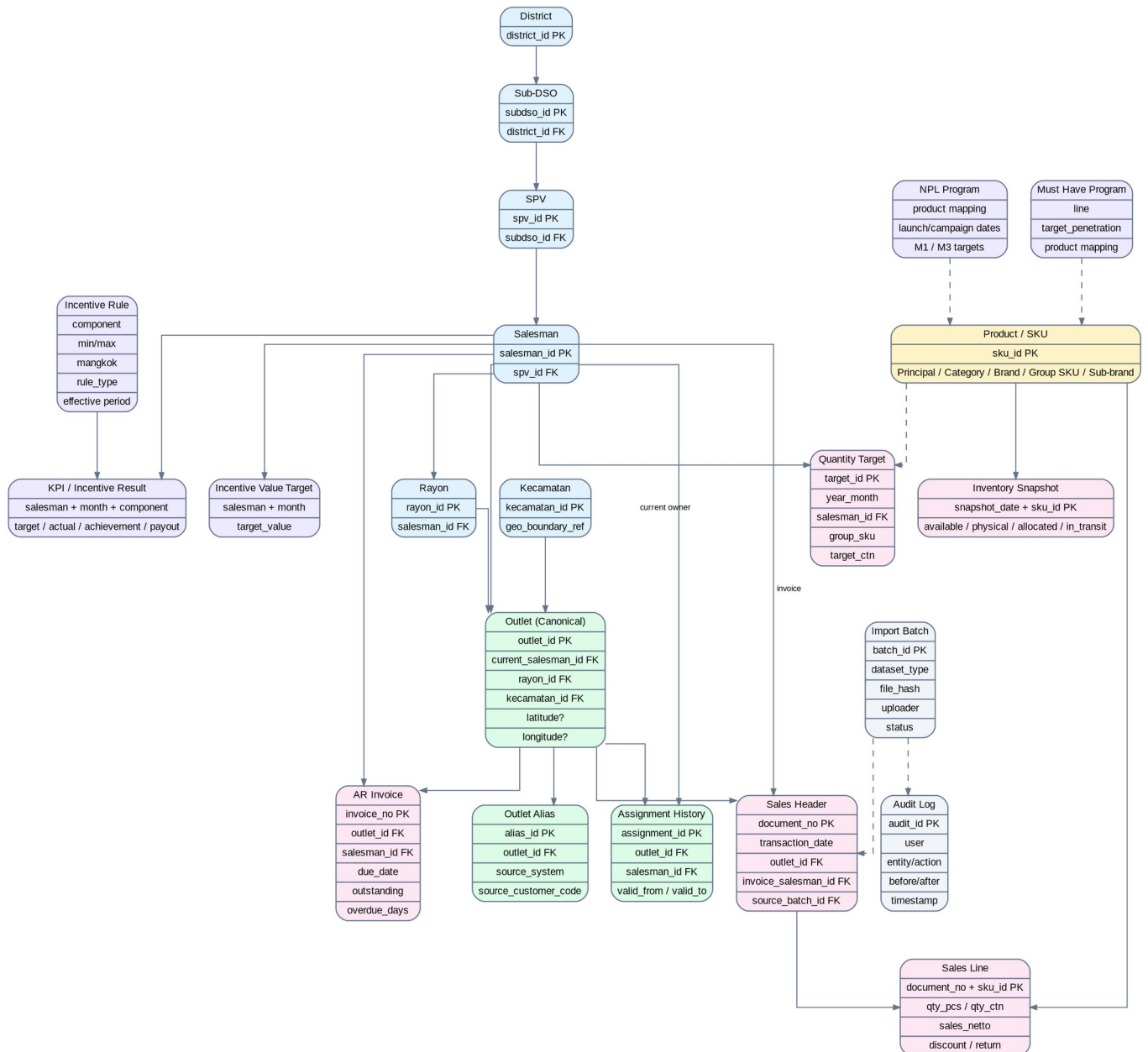


Figure 1. Conceptual ERD - Sales Garut Intelligence Platform

8.1 Key Modeling Decisions

- Canonical outlet identity is separate from source-system customer codes. Store source codes as aliases.
- Outlet assignment history is retained even though management reporting defaults to current ownership.
- Sales header preserves invoice salesman; management ownership is resolved separately.
- Quantity target and incentive value target are separate facts because they serve different business purposes.
- Inventory is a snapshot fact in V1; movement ledger is future scope.
- Latitude/longitude remain nullable in V1; visit/GPS facts are future scope.

- Import batches link business facts back to source files for traceability.

Entity	Purpose	Key	Important Note
dim_outlet	Canonical customer/outlet identity	outlet_id	Never use one raw source code as the permanent platform identity.
outlet_alias	Maps ERP/DSO/SFA/legacy codes to canonical outlet	alias_id	Supports multiple identifiers per physical outlet.
outlet_assignment_history	Salesman ownership history	assignment_id	Supports transfer rules and audit.
fact_sales_header	Document-level commercial event	document_no	Stores invoice salesman and canonical outlet.
fact_sales_line	SKU line within document	document_no + sku_id	Qty, value, returns.
fact_quantity_target	Carton target by month/salesman/group SKU	target_id	Primary target fact.
fact_incentive_value_target	Value target used only by incentive component	id	Separate from primary sales target.
fact_inventory_snapshot	Point-in-time stock by SKU	snapshot_date + sku_id	V1 inventory model.
fact_ar_invoice	Outstanding receivable invoice	invoice_no	Aging, outstanding, outlet/salesman scope.
incentive_rule	Configurable KPI/incentive rules	rule_id	Effective-dated thresholds and payout type.
import_batch	Source-file traceability	batch_id	Hash, uploader, status, timestamps.
audit_log	Changes to critical business data	audit_id	Before/after values where relevant.

9. Application & Data Architecture Requirements

Layer 1 - Raw Ingestion

Store original uploaded files immutably with upload metadata and file hash.

Layer 2 - Validation & Staging

Detect dataset type, normalize safe formatting, validate keys/dates/types, surface warnings/errors before commit.

Layer 3 - Canonical Storage

Normalized relational database containing organization, outlets, products, targets, sales, stock, AR, KPI/incentive and governance entities.

Layer 4 - Business Logic / Metrics

Centralized calculations for calendar pace, achievement, coverage, Must Have, NPL, dormancy, stock cover and incentive.

Layer 5 - Application / API

Role-aware web application, responsive dashboards, detail pages, import workflows and settings.

Layer 6 - Insight Layer

Deterministic rule-based insights and alerts with evidence links.

10. Module Specifications

Each module specification defines the product outcome, core content, user actions, filters/dependencies, and minimum acceptance criteria. Antigravity may improve layout and interaction patterns, but must not remove required business outcomes.

10.1 Beranda - Executive Command Center

Spec	Requirement
Primary users	DSM/Admin, SPV; limited salesman home variant
Purpose	Give a 30-second situational view of commercial performance, pace, coverage, product programs and immediate risks.
Core content	Target KTN; Actual KTN; Achievement; Remaining Target; Required Daily Pace; LE/LE%; HK/HKE/Remaining HK; Sales Value; Registered Outlet; Active Outlet; Coverage%; Dormant 60D; Must Have summary; NPL summary; stock warnings; AR warnings; team scorecard; trend/driver summaries.
Key actions	Change global filters; click KPI to open evidence/detail; navigate to salesman/product/outlet/area; view rule-based insights.
Filters / scope	Period, SPV, Salesman, Principal, Brand, Group SKU, Kecamatan.
Dependencies	Sales, target, calendar, CL/outlet, product mapping; stock/AR summaries when loaded.
Acceptance	Same scoped metrics reconcile with detail modules; no stale hard-coded HK; empty states explain missing source data.

10.2 Penjualan - Sales Performance

Spec	Requirement
Primary users	DSM/Admin, SPV, Salesman within permission scope
Purpose	Analyze what is driving sales performance across people, product, time, rayon and geography.
Core content	Cartons, pieces, Sales Netto, target cartons, achievement, gap, required pace, LE, contribution, MoM, YoY, YTD, returns ratio; product hierarchy drilldown; daily/weekly/monthly trends.
Key actions	Drill Principal / Category / Brand / Group SKU / Sub-brand -> SKU; compare periods; open salesman/outlet/area contributors; export tables.
Filters / scope	Period, SPV, Salesman, Principal, Brand, Group SKU, SKU, Rayon, Kecamatan.
Dependencies	Validated transaction grain, product hierarchy, target master, current ownership mapping.
Acceptance	Aggregations reconcile from SKU to total; target rows do not duplicate when product hierarchy filters change.

10.3 Outlet & Customer Intelligence

Spec	Requirement
Primary users	DSM/Admin, SPV, Salesman within scope
Purpose	Turn the customer list into an actionable coverage and retention system.
Core content	Registered, Active MTD, Coverage%, Dormant 60D, Never Active (within available history), coverage gap, Pareto/contribution, at-risk/dormant

	watchlist.
Key actions	Filter, search, sort contribution/recency, open Outlet 360, export outlet list.
Filters / scope	Period, SPV, Salesman, Rayon, Kecamatan, Principal/Brand for product coverage views.
Dependencies	Canonical outlet identity, CL status, current owner, sales history.
Acceptance	Inactive MTD is distinct from Dormant 60D; registered denominator follows current filter universe.

10.4 Garut Coverage Map

Spec	Requirement
Primary users	DSM/Admin, SPV; salesman scoped view
Purpose	Visualize outlet coverage gaps geographically before outlet-level coordinates are available.
Core content	Kecamatan choropleth by Coverage%; tooltip/card with Registered Outlet, Active Outlet, Coverage%, Sales KTN, Sales Value, active salesman count.
Key actions	Click kecamatan to drill into outlet list/performance; change period/team/product filters.
Filters / scope	Period, SPV, Salesman; optional Principal/Brand.
Dependencies	Validated Kecamatan naming and external validated Kecamatan boundary GeoJSON.
Acceptance	Coverage on map equals coverage in outlet module for the same scope; map does not infer boundaries from text addresses.

10.5 Outlet 360

Spec	Requirement
Primary users	All roles within scope
Purpose	Provide one complete, explainable view of each outlet.
Core content	Identity, all known source codes, address, Kecamatan, market, Rayon, current salesman, SPV, cluster, Active MTD, Dormant 60D, last order, days since last order, MTD/YTD cartons/value, order frequency, average order/drop size, contribution, trends, product mix, Must Have, NPL, credit/TOP, AR.
Key actions	Search/open outlet, inspect transactions, product mix, AR; admin-authorized mapping/assignment corrections with audit.
Filters / scope	Period and product context where applicable.
Dependencies	Canonical identity and aliases, sales, product, assignment, AR.
Acceptance	Current owner and invoice salesman are not conflated; original source IDs remain visible/auditable.

10.6 Tim Sales - Salesman Performance

Spec	Requirement
Primary users	DSM/Admin, SPV; salesman own view
Purpose	Connect performance, coverage, product execution and incentive progress per salesman.

Core content	Target/actual/achievement/gap/LE; sales value; growth; CL/OC/coverage/dormant; product performance; Rayon and Kecamatan breakdown; KPI component progress; incentive current/next thresholds.
Key actions	Rank/sort team; open salesman; drill product/outlet/area; inspect incentive component evidence.
Filters / scope	Period, SPV, Salesman, product, Rayon, Kecamatan.
Dependencies	Current ownership attribution, target, CL, product programs, incentive rules.
Acceptance	Management view attributes outlet history to current owner while audit view can show original invoice salesman.

10.7 Program Produk - Must Have

Spec	Requirement
Primary users	DSM/Admin, SPV, Salesman
Purpose	Measure outlet penetration against registered customer universe for Must Have lines.
Core content	KTG 45%, DELI 45%, RTD 30%, FOX 35%, UHT 40%; CL, Target OC, Actual OC, penetration, achievement, gap per line and total; salesman/SPV/DSO views.
Key actions	Drill gap outlets; inspect which CL outlets have/not purchased each line; configure program mapping/threshold if authorized.
Filters / scope	Period, SPV, Salesman, Rayon, Kecamatan, line.
Dependencies	Must Have product mapping, CL, sales.
Acceptance	Distinct outlet counting per line; same outlet may count in multiple lines; thresholds configurable.

10.8 Program Produk - NPL

Spec	Requirement
Primary users	DSM/Admin, SPV, Salesman
Purpose	Track launch penetration and repeat quality, not just initial distribution.
Core content	CL, NPL OC, penetration, M1/M3 target, first-order drop size, RO-1, RO-1%, RO-2+, RO-2+%, repeat drop size, movement.
Key actions	Select NPL campaign; drill salesman/outlet; configure campaign dates/targets if authorized.
Filters / scope	Campaign, period, SPV, Salesman, Rayon, Kecamatan.
Dependencies	NPL master, product mapping, sales history.
Acceptance	Order counts are campaign-scoped; RO-1 exactly 2 and RO-2+ >=3 orders.

10.9 KPI & Incentive

Spec	Requirement
Primary users	DSM/Admin, SPV, Salesman own view
Purpose	Make incentive progress transparent and actionable.
Core content	Six components; Mangkok; min/max; target; actual; achievement; current payout; gap to minimum; required per remaining HK; gap to maximum

	where applicable; total estimated incentive.
Key actions	Inspect calculation evidence; admin config rule effective dates; upload/manual input value target; export scorecard.
Filters / scope	Period, SPV, Salesman.
Dependencies	Quantity targets, sales, incentive value target, Must Have result, AR performance once formula confirmed, calendar.
Acceptance	Components 1-5 calculate using locked rules; AR incentive remains clearly unavailable until formula is configured - never guessed.

10.10 Stock & Stock Cover

Spec	Requirement
Primary users	DSM/Admin, SPV; salesman read-only if granted
Purpose	Read current SKU availability against target demand and actual sales velocity.
Core content	Available, physical, allocated (if source), in transit; target-based cover; actual-velocity cover; low/healthy/overstock status once thresholds configured.
Key actions	Filter/drill SKU/brand; inspect cover basis and rolling period; upload new snapshot.
Filters / scope	Snapshot date, Principal, Brand, Group SKU, SKU.
Dependencies	Inventory snapshot, product master, target, sales history.
Acceptance	Every cover metric labels its basis; current snapshot is not presented as movement history.

10.11 Piutang / AR & Credit

Spec	Requirement
Primary users	DSM/Admin, SPV; salesman own scope
Purpose	Expose credit risk alongside commercial performance.
Core content	Outstanding, overdue, aging buckets, oldest overdue days, by salesman/outlet, credit limit and TOP where available.
Key actions	Open outlet, filter aging, export follow-up list.
Filters / scope	As-of date, SPV, Salesman, Kecamatan, aging bucket.
Dependencies	Official AR export for production; outlet mapping.
Acceptance	AR amounts reconcile to import source; AR incentive formula is separated from operational AR reporting.

10.12 Data Center, Masters & Settings

Spec	Requirement
Primary users	DSM/Admin and explicitly permitted operators
Purpose	Allow safe self-service data updates without database access.
Core content	Excel dropzone; file detection; column mapping; validation preview; commit/rollback; import history; Customer/Product/Salesman/Territory masters; target input; incentive value target; program settings; working calendar; business settings; duplicate outlet review.

Key actions	Upload, dry-run, review errors, commit, rollback/reverse, edit authorized masters, merge/unmerge identity mappings, configure rules.
Filters / scope	Dataset type, date, uploader, status.
Dependencies	Raw file storage, staging, validation engine, RBAC, audit log.
Acceptance	Repeated identical file cannot silently duplicate facts; failed imports leave production data intact; critical changes are audited.

11. Rule-Based Insight Engine

V1 should use deterministic and explainable insights, not opaque AI/ML. Each insight must contain the trigger, supporting metric/context, and a click path to evidence.

Insight Type	Example
Behind Pace	Achievement 46% while 72% of working days have elapsed.
Coverage Opportunity	Kecamatan X has 210 registered outlets and 102 active outlets.
Dormancy	Outlet ABC has had no purchase for 60 days.
Product Driver	Brand X explains a material share of current growth.
Must Have Gap	DELI penetration is 31% vs target 45%.
Stock Risk	SKU X has 2.4 days actual-velocity cover.
AR Risk	Overdue >30 days is concentrated in a small outlet set.

12. UX / UI Requirements

- Default UI language: Bahasa Indonesia.
- Use business labels; never expose database/table names to normal users.
- Clean, premium, restrained visual style with strong hierarchy and comfortable spacing.
- Do not overload one page with charts. Prefer KPI cards, one or two primary visuals, concise tables and progressive drilldown.
- Filters must remain consistent and understandable. Preserve selected context when navigating to detail where sensible.
- Desktop-first analytics for DSM/SPV; fully responsive mobile experience for salesman viewing.
- Empty/error states must explain what data is missing and how to resolve it.
- Negative Excel-style gaps should be expressed as actions when possible (for example: Need 56 KTN to reach minimum).

13. Data Import, Quality, Security & Audit

13.1 Import Workflow

Drop Excel -> Detect -> Map -> Dry Run -> Validate -> Review -> Commit -> Audit

- Supported V1 datasets: Sales Transactions, Customer List, Quantity Target, Incentive Value Target, Stock Snapshot, AR.
- Detect duplicates, invalid dates, unknown outlet/SKU/salesman, missing required fields, corrupted/scientific notation IDs, and inconsistent mapping.
- Original file is immutable. Store upload ID, filename, uploader, timestamp, file hash and status.
- Import must be idempotent or provide a robust duplicate-prevention strategy.
- Rollback/reversal must be possible without deleting the raw source.

13.2 Security

- Authentication and secure sessions
- Server-side authorization on every protected request
- Role/capability scoped data access
- Salesman cannot access another scope by changing URL/API parameters
- Restricted admin operations
- Audit log for critical changes

13.3 Audit / Traceability

Critical data including target, outlet assignment, customer mapping, incentive rule, Must Have threshold and working calendar must have change history. Users should be able to answer: "Where did this number come from?" through source batch and calculation context.

14. V1 Scope & Product Acceptance Criteria

Capability	V1 Status
Authentication & RBAC	Required
Excel Import Center & Validation	Required
Customer/Product/Salesman/Rayon/Kecamatan masters	Required
Quantity Target	Required
Incentive Value Target input/upload	Required
Executive Command Center	Required
Sales Performance	Required
Salesman Performance	Required
Outlet 360	Required
Coverage & Dormancy	Required
Garut Kecamatan Map	Required
Must Have Tracker	Required
NPL Tracker	Required
KPI & Incentive components 1-5	Required
AR operational monitor	Required
AR incentive	Blocked until formula confirmed
Stock Snapshot & Stock Cover	Required
Rule-Based Insights	Required
Audit Trail	Required
GPS Visit Tracking	Not V1

14.1 Cross-Module Acceptance

- Same metric + same filters = same number across Home, Sales, Salesman, Outlet and Map modules.
- Coverage correctly separates Active MTD from Dormant 60D.
- Current-owner attribution is used by default for management views while invoice salesman remains auditable.
- Map coverage exactly reconciles with tabular kecamatan coverage.
- Must Have distinct-outlet logic is consistent across salesman, SPV and DSO aggregation.

- AR incentive is not fabricated when formula is unavailable.
- No production claim is made until full transaction export passes UAT.

15. Implementation Milestones

Milestones are outcome-based. Antigravity may choose the exact technical stack and internal sprint size, but each milestone must meet its exit criteria before the next business layer is considered complete.

Milestone	Scope	Exit Criteria	Guardrail
M0 - Technical Foundation & Data Contracts	Establish repository/app shell, environment strategy, relational database, authentication skeleton, canonical entity contracts, raw/staging conventions and test fixtures.	ERD implemented at schema level; migration framework; role model; source-data contracts; automated tests for core IDs/grains.	No dashboard dependency on raw Excel files.
M1 - Data Center & Canonical Masters	Build raw upload storage, Excel parsing, auto-detection/mapping, dry-run validation, commit workflow, import history, Customer/Product/Salesman/Rayon/Kecamatan masters, canonical outlet aliases and duplicate review.	At least each V1 dataset can be staged; invalid rows are visible; repeated imports do not silently duplicate; audit records created.	A bad file cannot corrupt committed data.
M2 - Commercial Metrics Engine	Implement working calendar, target/actual, achievement, gap, required pace, LE, growth, returns, current-owner attribution and aggregation services.	Unit tests for formulas; reconciliation test against validated examples; configurable HK/holidays.	Metrics are centralized, not reimplemented per screen.
M3 - Executive & Sales Performance	Build global filters, Executive Command Center, Sales Performance drilldown and trend analysis.	DSM can answer current performance, pace and key product/team drivers; metrics reconcile across modules.	No hard-coded sample totals.
M4 - Customer Intelligence, Outlet 360 & Coverage Map	Build CL/OC, coverage, Dormant 60D, Outlet 360, contribution/Pareto, watchlists and Garut kecamatan choropleth.	Coverage reconciles by DSO/salesman/kecamatan; map drilldown works; outlet current-owner/alias identity is clear.	Validated GeoJSON required for map.
M5 - Must Have & NPL Programs	Build configurable Must Have mappings/thresholds and NPL campaign engine with repeat logic.	KTG/DELI/RTD/FOX/UHT targets work by salesman and DSO; NPL penetration/RO metrics reconcile with test cases.	Program definitions live in data/config, not code constants.
M6 - Sales Force KPI & Incentive	Build salesman scorecards and incentive components 1-5; value target input; incentive pace/action wording.	Component rules, min/max, payout caps, Must Have binary payout and total estimated incentive are testable and explainable.	AR incentive remains disabled/TBD until formula is provided.
M7 - Stock, AR & Insight Layer	Build stock snapshot/dual cover, AR aging/credit views and deterministic insights.	Stock cover clearly labels target vs actual basis; AR reconciles to source; insights deep-link to evidence.	No inventory movement claims from snapshot-only data.
M8 - Full-Data UAT & Production Readiness	Load full unsampled transaction extract, resolve mappings, run reconciliation, permission tests, performance tests and user UAT.	Signed reconciliation for agreed key metrics; no critical data-quality blockers; role access verified; production backup/restore and import rollback proven.	This milestone is the production gate.

16. Open Items & Explicit Non-Assumptions

Open Item	Required Treatment
Exact AR Performance formula	Must be confirmed before AR incentive payout is activated. Operational AR analytics can ship earlier.
Exact product membership for Kopi / Beverage / Non Kopi & Non Bvg	Create configurable product-group mapping and validate against business master during build/UAT.
Official source for Incentive Value Target	Support manual input/dedicated upload until source is confirmed.
Returns treatment in target attainment	Validate against approved business reporting; do not assume from corrupted spreadsheet cells.
Garut Kecamatan boundaries	Use a validated GeoJSON/boundary source; do not approximate

	boundaries.
Full transaction extract	Mandatory for final production UAT.
Future NPL campaign dates	Store as configurable campaign master fields.
GPS/geofence rules	Not required in V1; no geofence threshold should be embedded now.

17. Roadmap Beyond V1

17.1 V1.5 - Operational Enhancement

- Alert center and follow-up/task list
- Improved outlet duplicate-resolution workflow
- Stock snapshot history and trends
- Enhanced salesman mobile UX
- Outlet latitude/longitude collection
- Point-map visualization
- Visit planning foundation

17.2 V2+ - Sales Force Automation & Advanced Intelligence

- Check-in/check-out
- Event-based location capture
- Visit result and effective calls
- Visit-to-order conversion
- Offline field activity
- Route analysis and optimization
- ERP/API sync
- Daily stock feeds and replenishment signals
- Forecasting/churn prediction
- Recommended outlet/product actions
- Multi-district deployment

18. Final Product Definition

Sales Garut Intelligence Platform V1 is an operational sales intelligence platform that lets DSM, SPV and Salesman understand target, sales, outlet coverage, product performance, territory, KPI, incentive, stock and AR through one governed source of truth.

The product must answer three levels of questions:

Level	Question	Expected Experience
WHAT?	What is happening?	Achievement 72%. Coverage 61%.
WHY?	Why is it happening?	A product line is behind, specific kecamatan coverage is low, and key outlets have not ordered.
WHAT NEXT?	What needs attention?	Need X KTN/day, Y additional Must Have outlets, and Z dormant outlets to reactivate.

Appendix A - Metric & Calculation Notes

Metric	Definition
HK	Total configured operating days in month.
HKE	Operating days elapsed through the selected monitoring date.
Time Rate	HKE / HK .
Remaining HK	$HK - HKE$.
Achievement	Actual / Target.
Required Daily Pace	$\max(\text{Target} - \text{Actual}, 0) / \text{Remaining HK}$.
LE	$\text{Actual} / HKE \times HK$.
Coverage	Distinct active CL outlets / registered CL outlets.
Dormant	No purchase ≥ 60 consecutive days.
Must Have Target OC	CL x line penetration target.
Must Have Actual OC	Distinct CL outlets purchasing mapped line in selected month.
RO-1	Exactly 2 NPL campaign orders.
RO-2+	3 or more NPL campaign orders.
Stock Cover Target	$\text{Available Stock} / (\text{Monthly Target} / HK)$.
Stock Cover Actual	$\text{Available Stock} / \text{configured rolling average daily actual sales}$.

Appendix B - Source & Validation Notes

This specification was derived from: (1) the original Sales Garut PRD / brainstorm; (2) Antigravity Discovery & Data Architecture Report, September 2026; (3) validated business-owner answers covering target unit, hierarchy, dormancy, reassignment, incentive structure, Must Have penetration targets, map direction and GPS deferral; and (4) the incentive sheet shared during validation.

Where source data or business definitions remain insufficient, this PRD explicitly labels the item TBD instead of silently inferring a rule. This is intentional and part of the product governance model.